Symmetry and skew

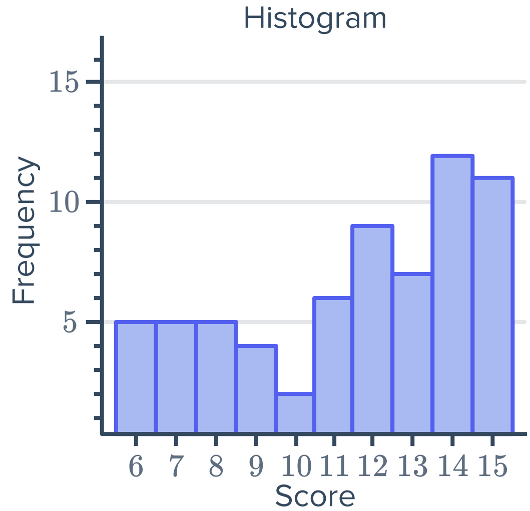
1 Describe the shape of the data in the following graphs:

a

	Leaf
1	6 7 7
2	2 2 2 2 3 3 3
3	3 3 3 6 6 6 7 7 7 7 7
4	4 4 4 4 4 4
5	7 7

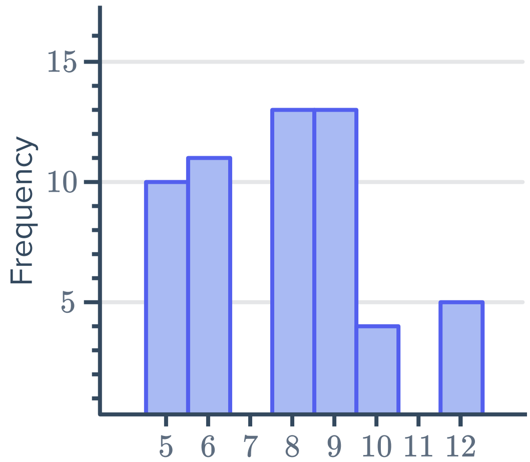
Key: $2|3 = 23$

b

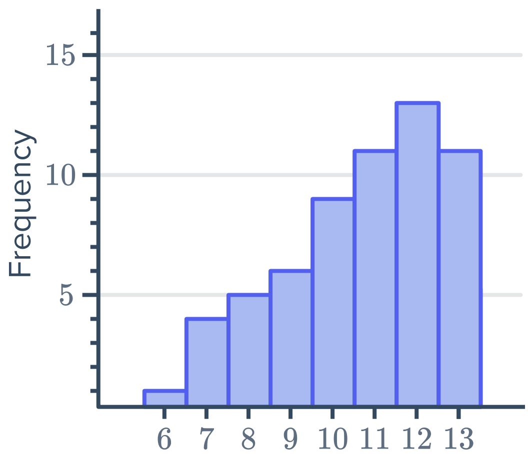


c

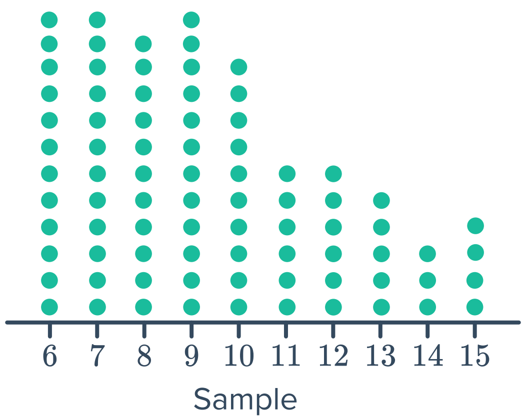
d



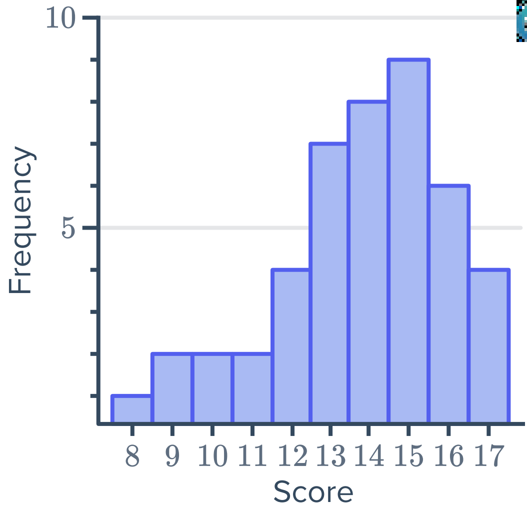
e



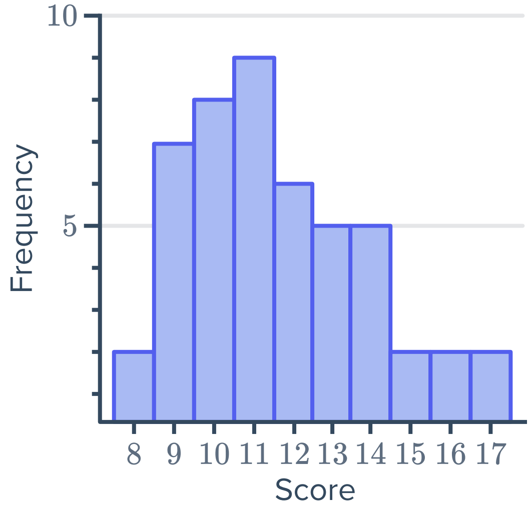
f



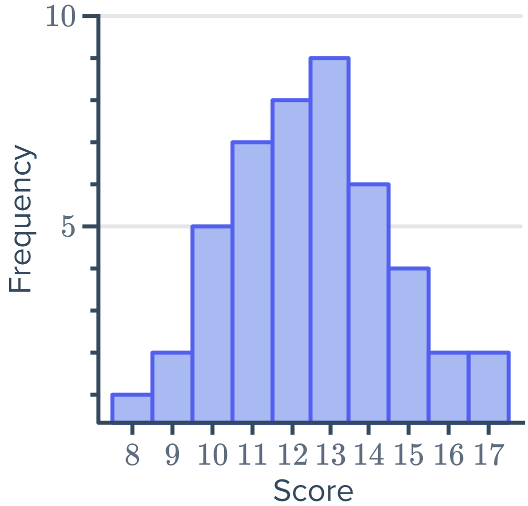
g


10
5
Frequency
8 9 10 11 12 13 14 15 16 17
Score

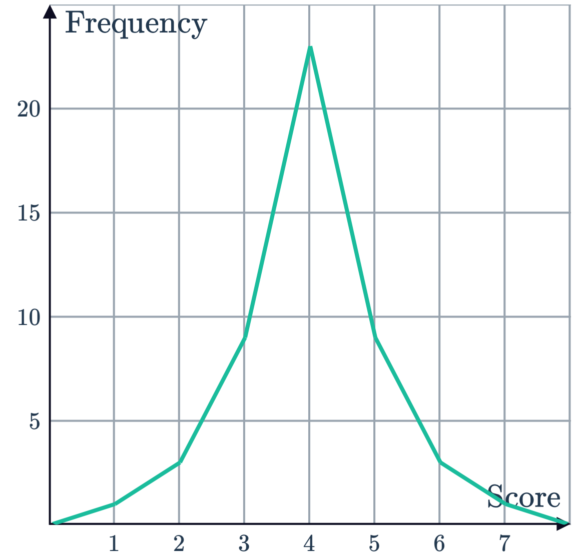
h


10
5
Frequency
8 9 10 11 12 13 14 15 16 17
Score

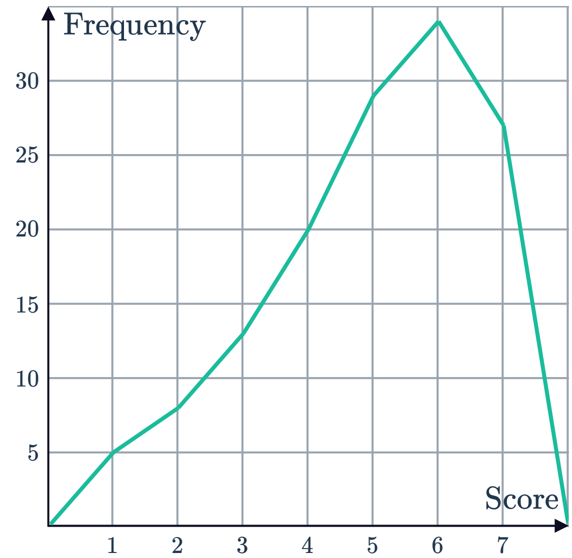
i


10
5
Frequency
8 9 10 11 12 13 14 15 16 17
Score

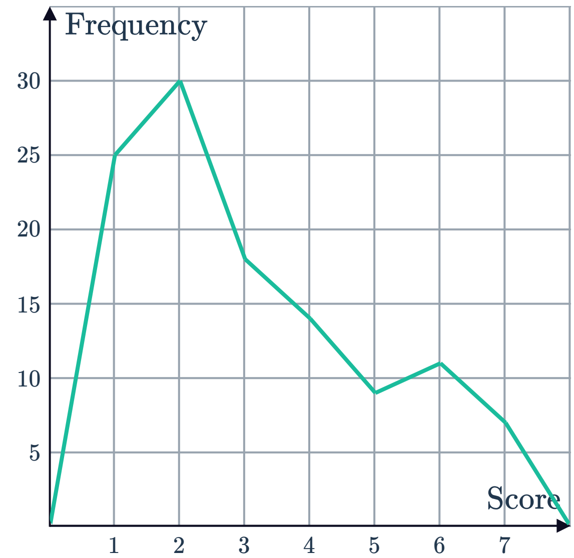
j


Frequency
20
15
10
5
Score
1 2 3 4 5 6 7

k

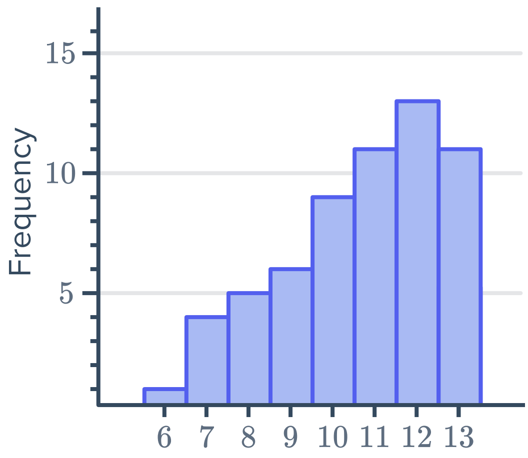

Frequency
30
25
20
15
10
5
Score
1 2 3 4 5 6 7

l

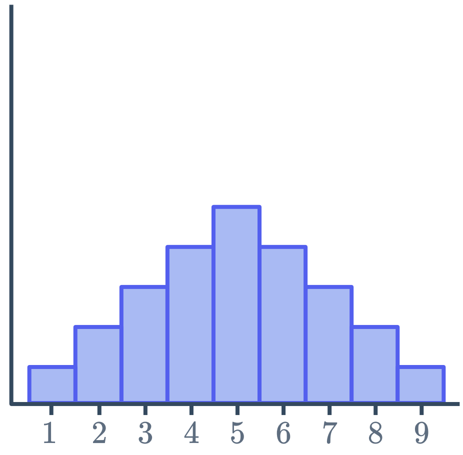

Frequency
30
25
20
15
10
5
Score
1 2 3 4 5 6 7

2 Determine whether the following graphs a[redacted]modal:

a



b



c

	Leaf
1	0 0 2 7
2	2 2 3 3 5 8
3	1 6 4
4	7
5	0 1 6
6	5 7 7 8 8
7	4 4 4
8	4 4

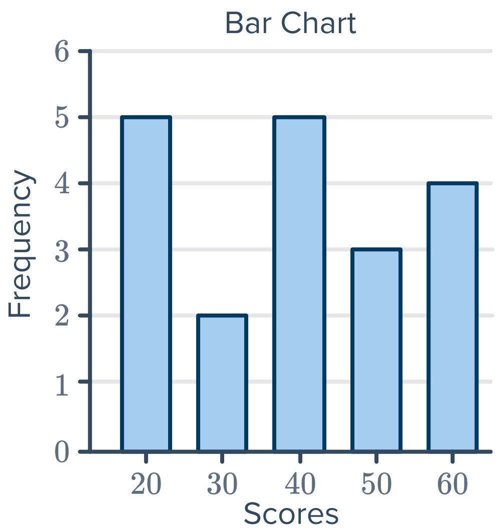
Key: $2|3 = 23$

d

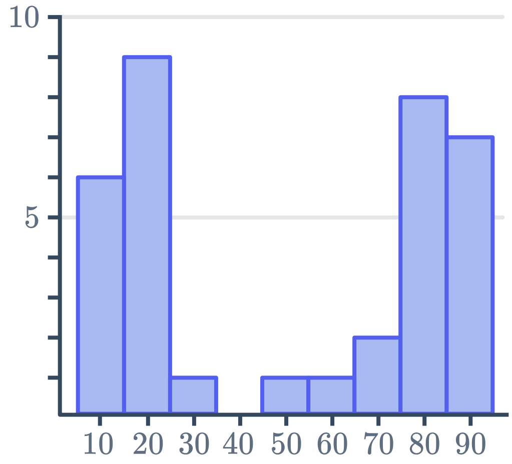
	Leaf
1	2 3 4 5 5 6 6 7 7 9
2	1 2 4 4
3	0 2 9 9
4	
5	5

Key: $1|2 = 12$

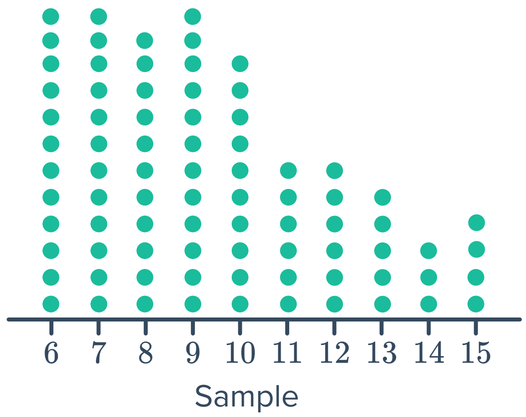
e



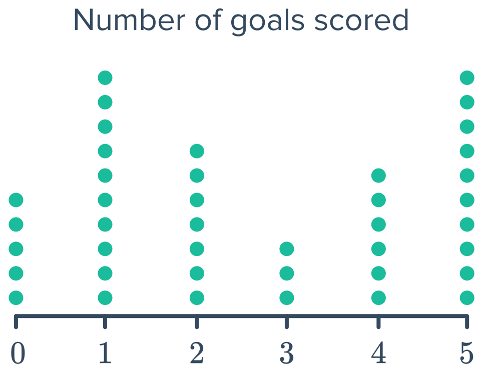
f



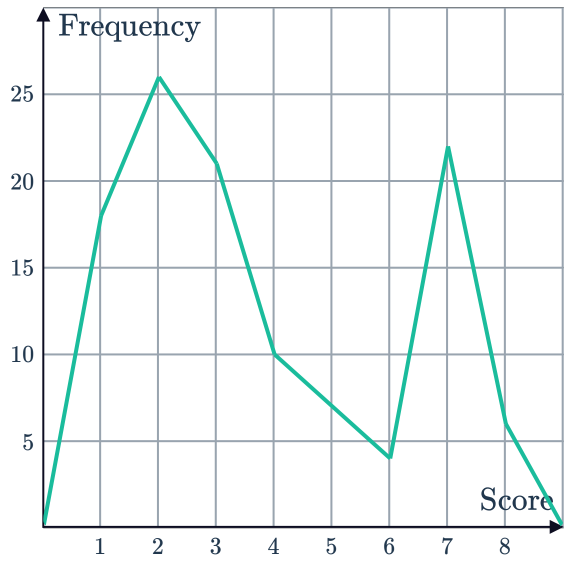
g



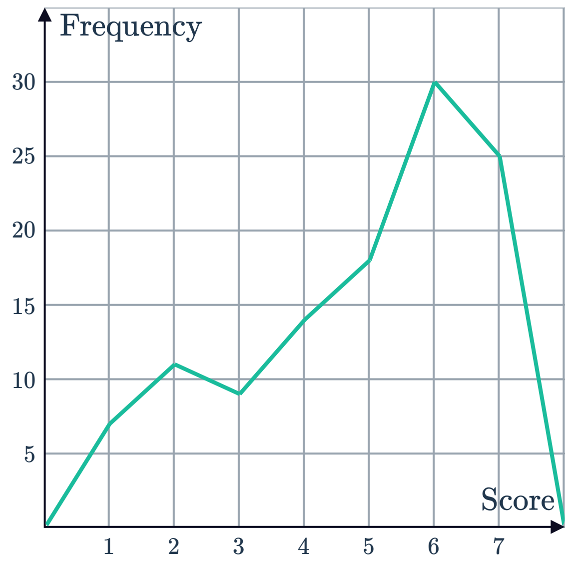
h



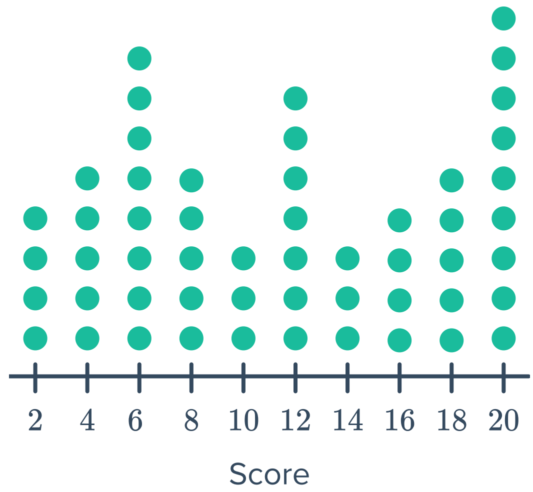
i



j



3 How would you describe the modality of the following dot plot? Explain your answer.



4 The table shows the number of crime novels in a bookshop for different price ranges rounded off to the nearest $\$5$:

a Graph this data as a histogram.

b Describe the shape of the distribution of the data.

Price of crime novel	Frequency
$\$5$	5
$\$10$	10
$\$15$	17
$\$20$	8
$\$25$	17
$\$30$	10
$\$35$	5

Clusters and outliers

5 The number of hours worked per week by a group of people is represented in the following stem-and-leaf plot:

a State the value of any outliers.

b Is there any clustering of data? If so, in what interval?

c State the mode.

	Leaf
0	2
1	
2	0 3 6 6
3	1 4 5 6 6 7
4	0 4 6 7 9
5	0

Key: $2|3 = 23$

6 Consider the stem plot given:

a State the value of any outliers.

b Is there any clustering of data? If so, in what interval?

c State the mode.

d Describe the shape of the distribution.

	Leaf
0	5
1	7 8
2	0 8
3	1 3 3 7 8 9
4	1 3 5 8 8 8
5	
6	
7	
8	
9	2

Key: $2|3 = 23$

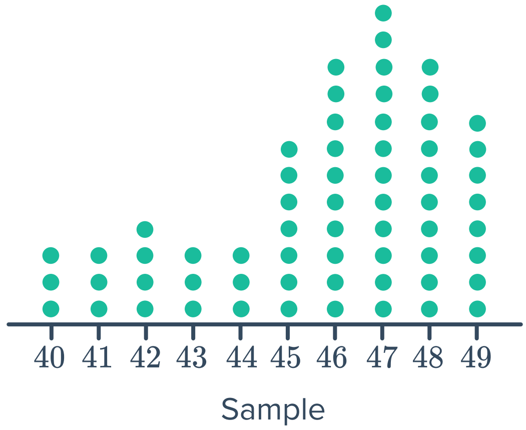
7 Temperatures were recorded over a period of time and presented as a dot plot:

a Are there any outliers?

b Is there any clustering of data? If so, in what interval?

c What is the modal temperature?

d Describe the shape of the distribution.



8 The number of peanuts in mixed nut packets were sampled and recorded in the following stem plot:

a Complete the frequency distribution table:

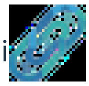
Score	Frequency
$40-49$	
$50-59$	
$60-69$	
$70-79$	
$80-89$	
$90-99$	
$100-109$	
$110-119$	

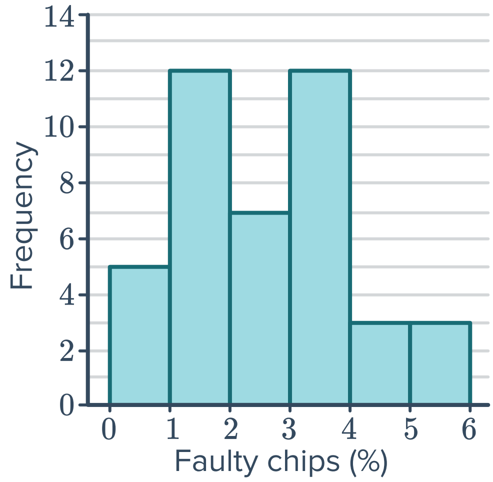
	Leaf
4	3 6 8
5	1 2 2 6 7 7 8
6	0 0 2
7	3 3 4 5 9
8	1 1 1 4 6 8 8 9
9	0 2 5 6
10	1 2 3 5 5 6 7 8
11	0 4 5 7

Key: $5|2=52$

b Describe the modaility of the distribution.

c State the modal class or classes of the data.

9 The percentage of faulty computer chips i
42 batches were recorded in the given histogram:



a Describe the modaility of the distribution.

b State the modal class or classes of the data.

10 Consider the dot plot below:

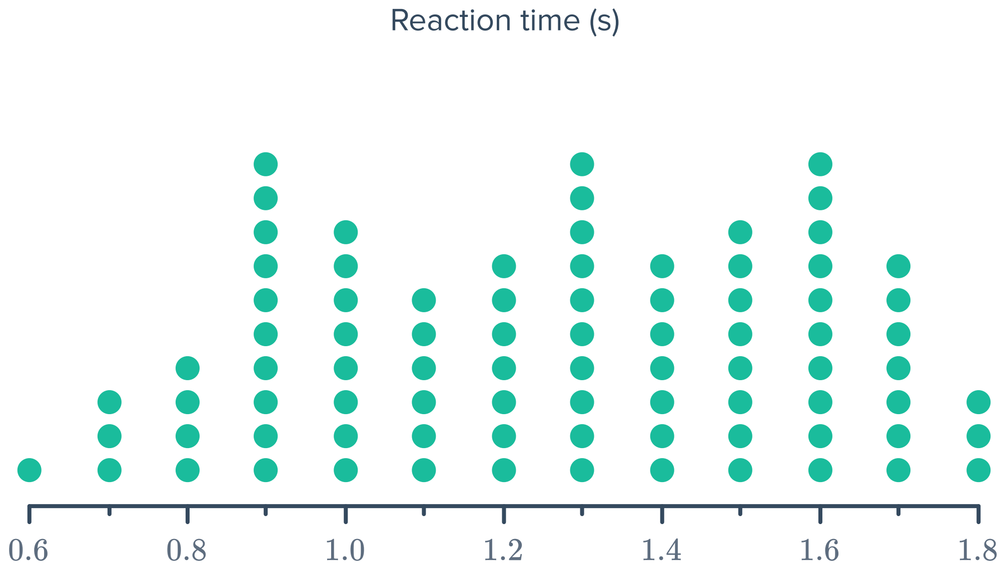
a Are there any outliers?

b Is there any clustering of data?

c State the modal score(s).

d Describe the shape of the distribution.

11 The reaction time of drivers was tested an[redacted]orded in the dot plot below:



a Construct a frequency distribution table for the individual data values.

b Describe the modality of the distribution.

c State the mode(s).

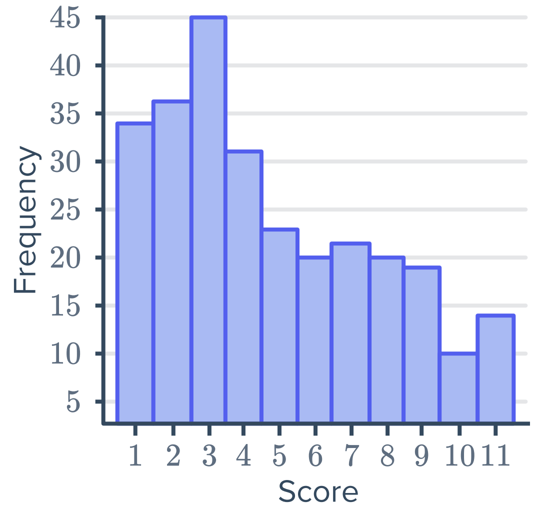
12 Consider the data shown in the histogram:

a Are there any outliers?

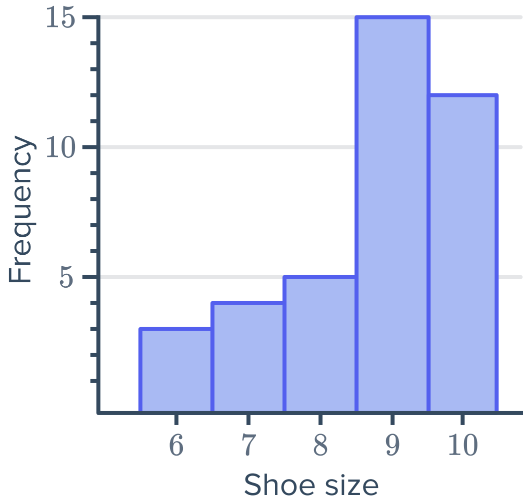
b Is there any clustering of data? If so, in what interval?

c State the mode.

d Describe the shape of the distribution.

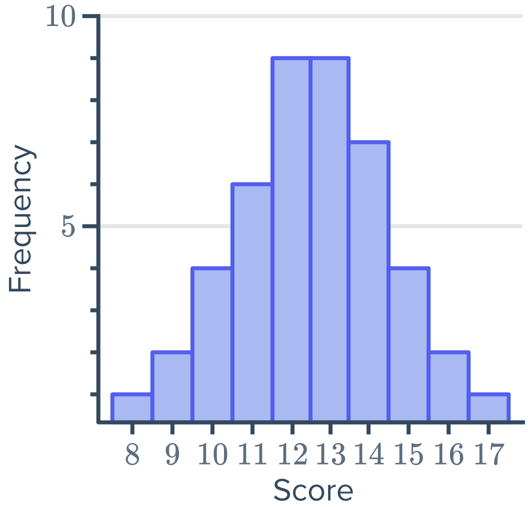


13 The shoe sizes of all the students in a class were measured and the data was presented in a graph:

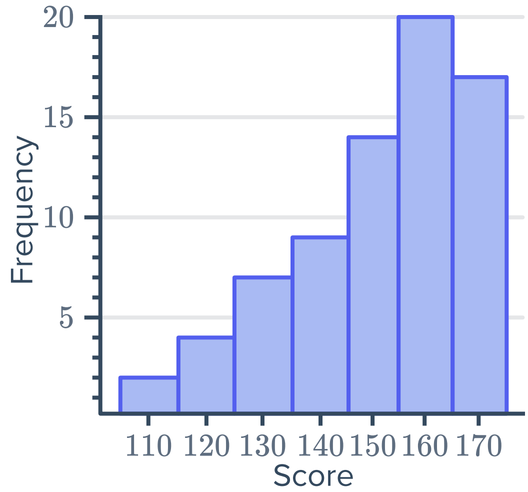


- **a** Are there any outliers?
- **b** Is there any clustering of data? If so, in what interval?
- **c** What is the modal shoe size?
- **d** Describe the shape of the distribution.

14 Estimate the value of the mean of the following data set correct to one decimal place:

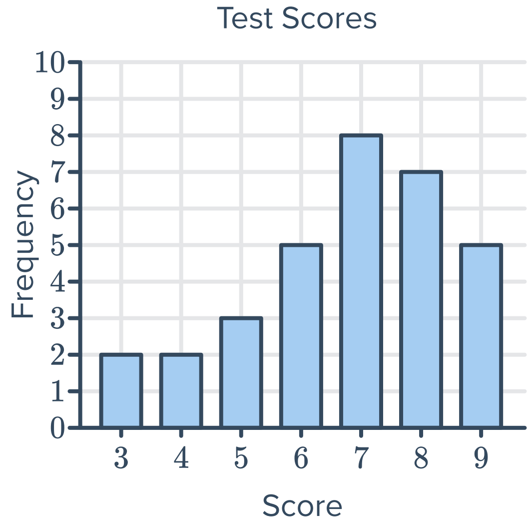


15 Consider the histogram representing students' heights in centimetres:



- **a** Does the histogram most likely represent grouped data or individual scores?
- **b** Estimate the value of the mean to one decimal place.
- **c** Describe the shape of the distribution.

16 Consider the given column graph:



a Describe the shape of the distribution.

b Find the following:

i Lower quartile

ii Upper quartile

c Hence, calculate the interquartile range.

d Are there any outliers? If so, state the value.

17 Consider the dot plot given:

a Describe the shape of the distribution.

b Find the following:

i Lower quartile

ii Upper quartile

c Hence, calculate the interquartile range.

d Are there any outliers? If so, state the value.

18 The stem-and-leaf plot below shows the age of people to enter through the gates of a concert in the first 5 seconds:

a Find the median age.

b Find the difference between the lowest age and the median.

c Find the difference between the highest age and the median.

d Calculate the mean age, correct to two decimal places.

e Is the data positively or negatively skewed?

	Leaf
1	0 1 2 2 3 3 4 4 4 8 8 8
2	1 7
3	4 5 5
4	0
5	4

Key: $1|2 = 12$ years old

19 $\mathrm{VO_2Max}$ is a measure of how efficiently y[redacted]dy uses oxygen during exercise. The more physically fit you are, the higher your $\mathrm{VO_2Max}$. Here are some people's results, listed in ascending order, when their $\mathrm{VO_2Max}$ was measured:

$$21, 21, 23, 25, 26, 27, 28, 29, 29, 29, 30, 30, 32, 38, 38, 42, 43, 44, 48, 50, 76$$

a Determine the median $\mathrm{VO_2Max}$.

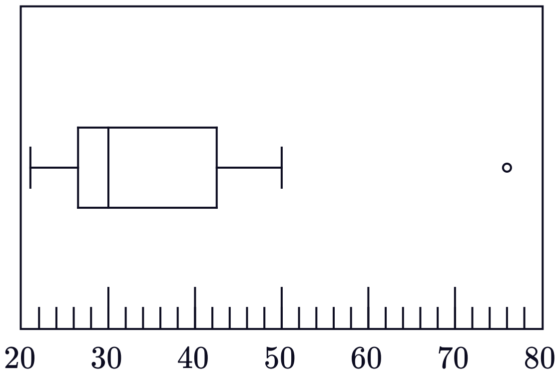
b Determine the upper quartile value.

c Determine the lower quartile value.

d Consider the box plot for this data set and state whether the results are positively or negatively skewed.

e State the value of the outlier.

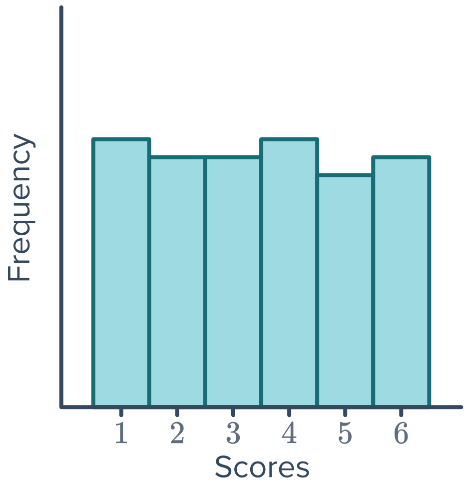
f An average untrained healthy person has a $\mathrm{VO_2Max}$ between 30 and 40. What can we say about the exercise habits of the majority of this group of people?



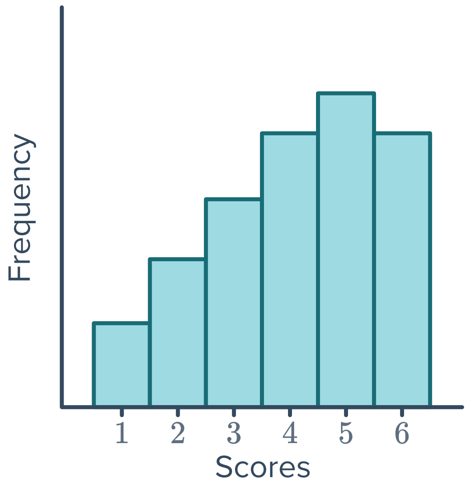
20 A die is rolled for a large number of trials and the number appearing is noted.

a Which histogram would you expect to match the data? Explain your answer.

A



B



b Describe the shape of the distribution of the data chosen above.

21 A pair of dice are rolled and the numbers appearing on the uppermost face are added to create a score.

a How many combinations would result in a score of:

i 4 **ii** 7 **iii** 10

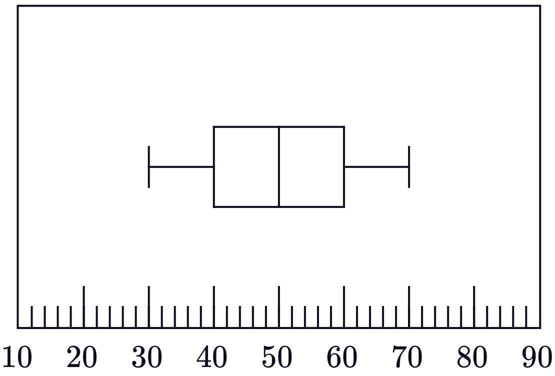
b A pair of dice are rolled for a large number of trials and the numbers appearing are added to create a score. Draw a sketch of a histogram you would expect to match the data.

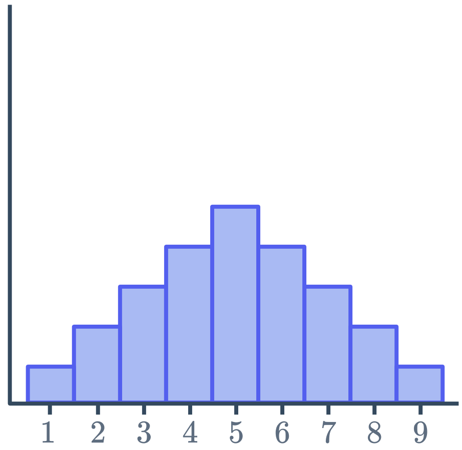
Connect histograms and box plots

22 Match the histograms on the left to the corresponding box plots on the right:

Histogram A

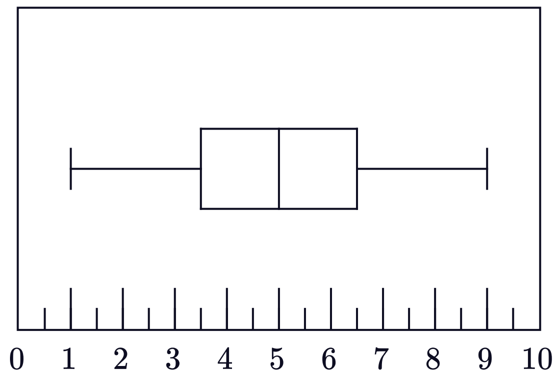
Box Plot 1

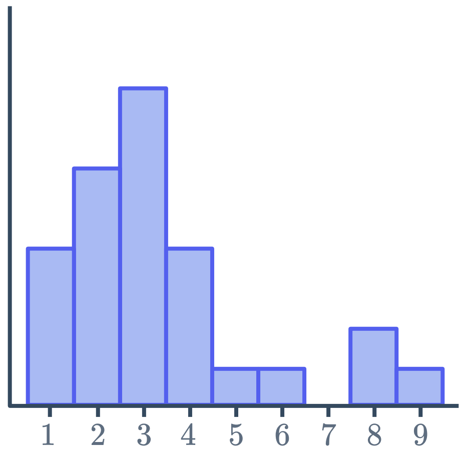




Histogram B

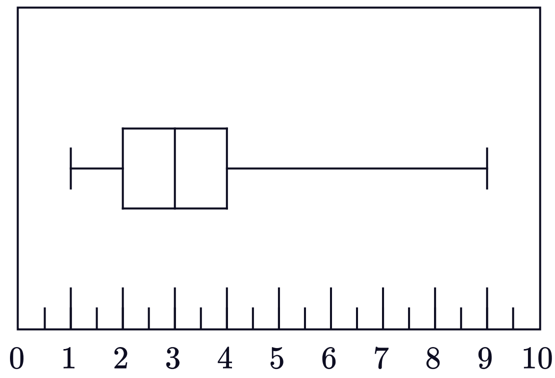
Box Plot 2

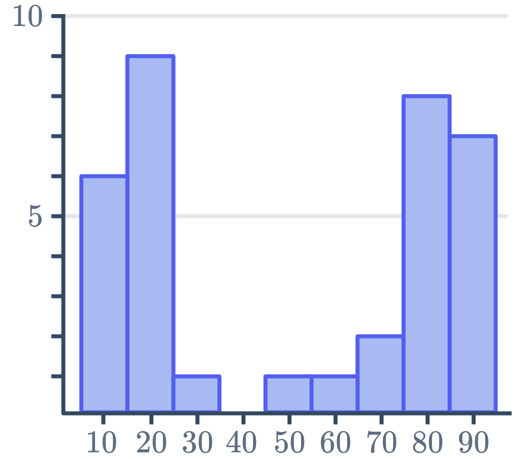




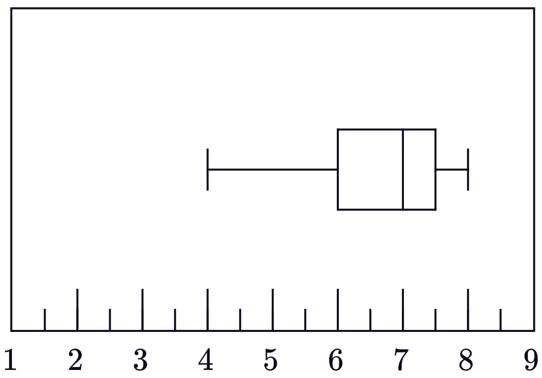
Histogram C

Box Plot 3

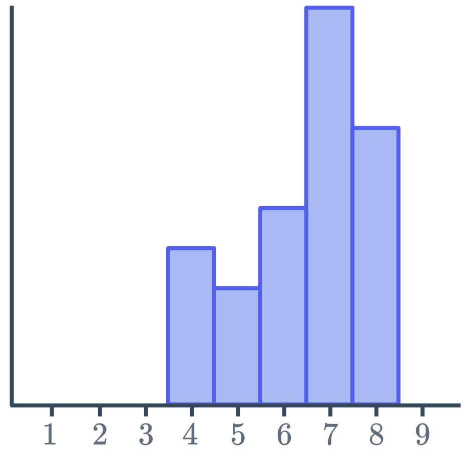




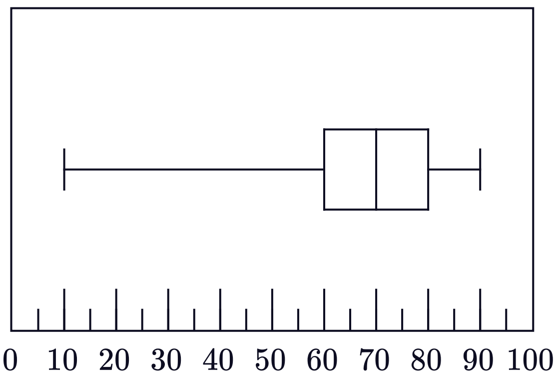
Box Plot 4



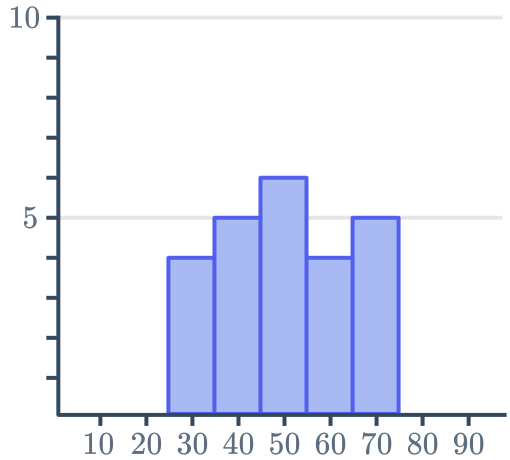
Histogram D



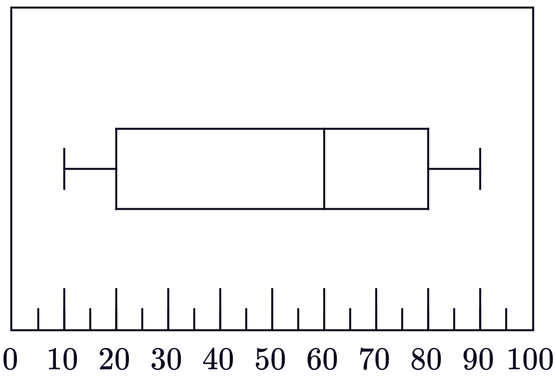
Box Plot 5



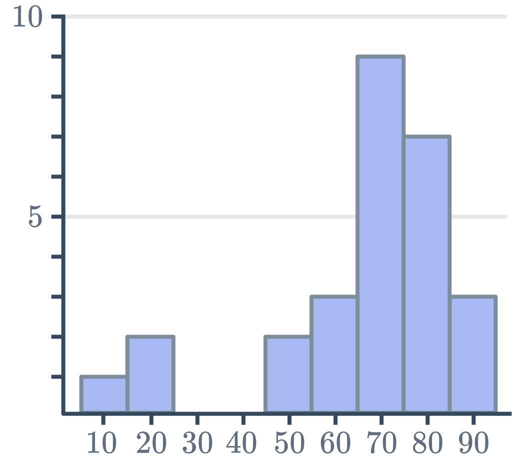
Histogram E



Box Plot 6

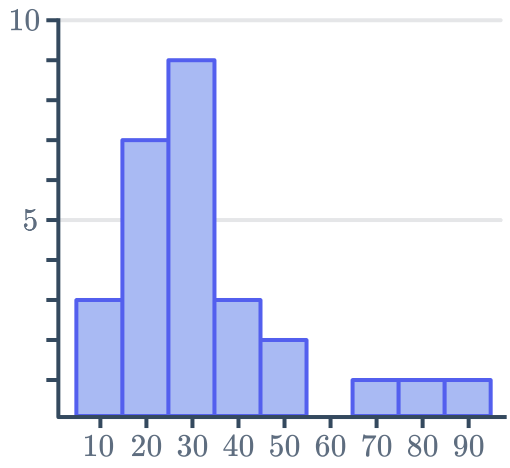


Histogram F

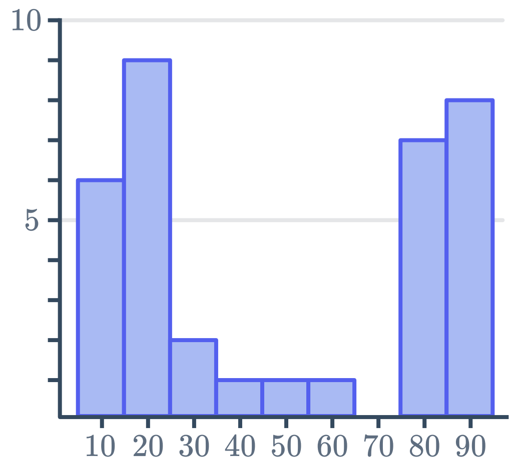


23 Construct a box plot for the following histograms:

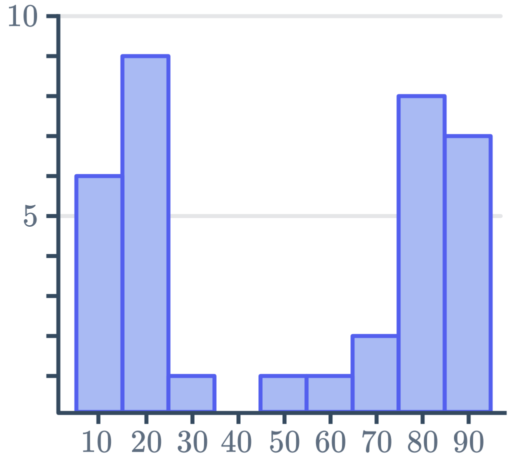
a



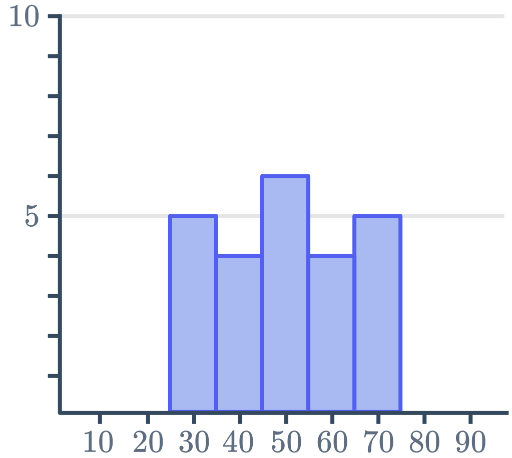
b



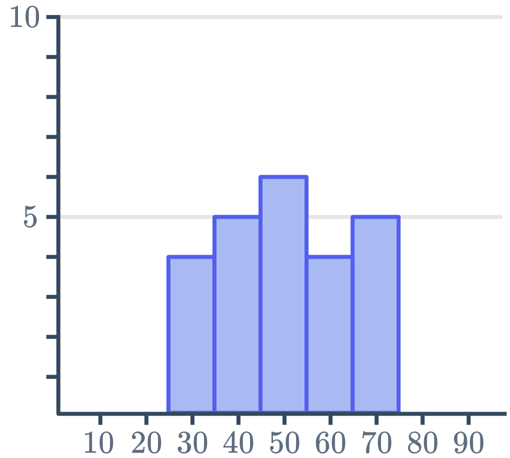
c



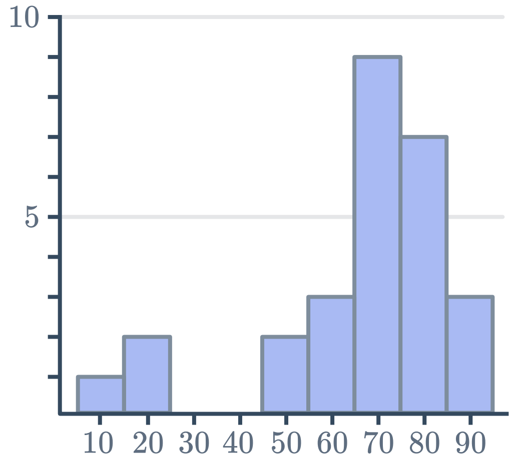
d



e

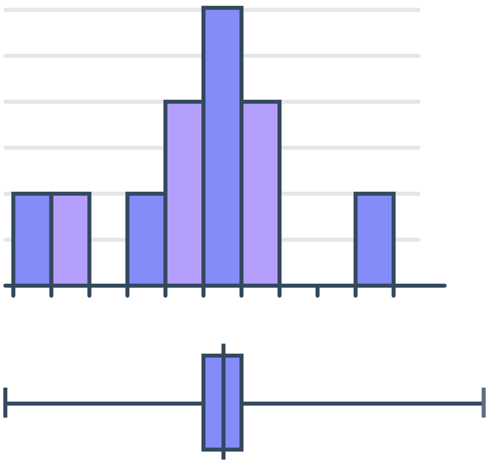


f

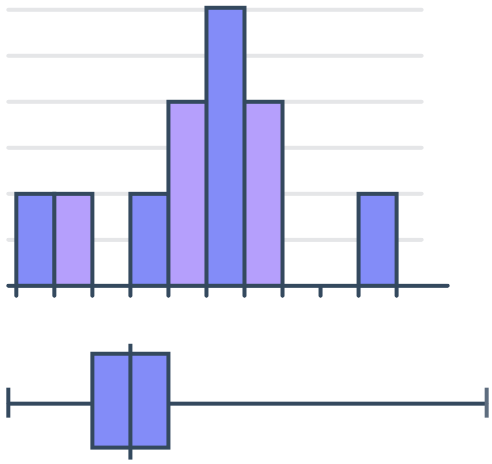


24 State whether the following pairs of histog[illegible] and box plots are correctly matched with respect to their shape:

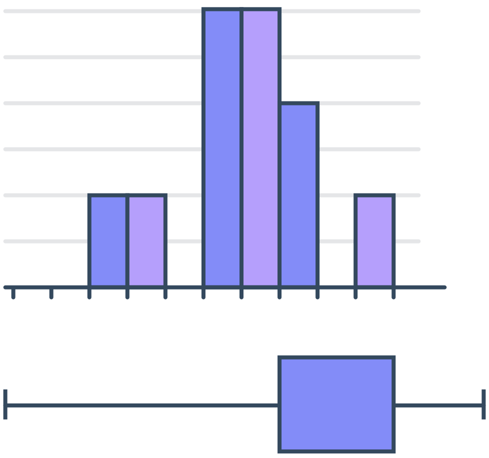
a



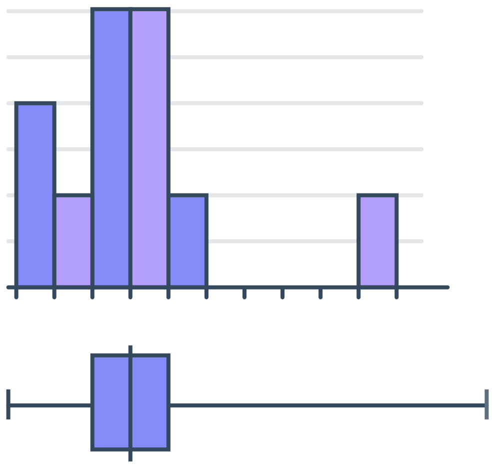
b



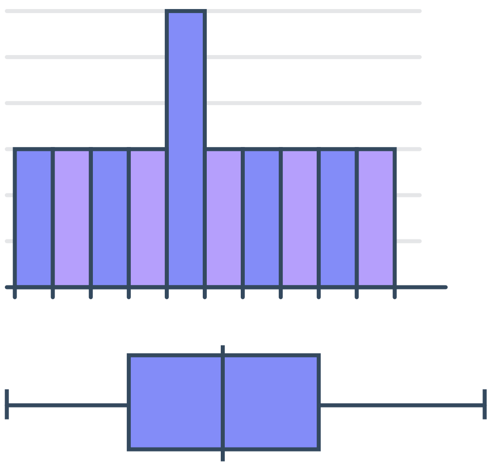
c



d

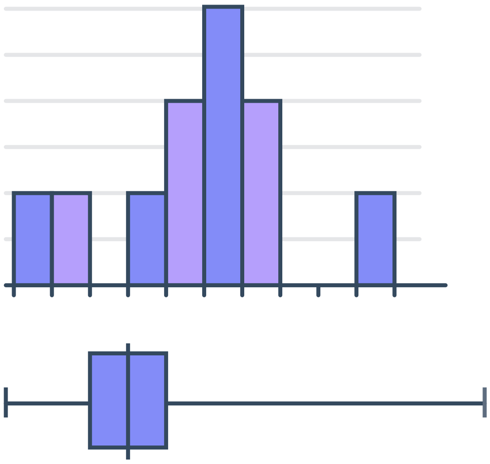


e



25 Explain why the following pairs of histograms and box plots do not match:

a



b